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Digitala trygghetslarm – Internetprotokoll för digitala trygghetslarm (SCAIP) – Specifikation

Digital social alarm – Social care alarm internet protocol (SCAIP) – Specification

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Introduction

This is a standard for an open IP-based communication protocol that can be used by social care services to support people living in their own homes or grouped living. Standardization of an open communication protocol is needed to ensure interoperability in the market of social care services.

The protocol defined in this standard will allow an open transparent non-proprietary information transfer and communication between users and social services via mainstream communication networks. An official standardised open protocol prevents social care alarms from being blocked or restricted in the European and global telecommunications networks and internet connections.

The protocol is defined to handle initiation, addressing, and transport functions based on SIP (Session Initiation Protocol) in order to set up a media stream and to transfer information between the user and the receiver. The data exchange of the protocol is defined as an XML schema including the alarm types, codes and additional information.

In this standard two generic use cases are described: one without establishment of a voice or multimedia session, and one use case describing an event message exchange combined with an additional voice or multimedia session using SIP.

The series of publications for social care alarms include the following:

SS 91100:2014	*Digital social alarm – Social care alarm internet protocol (SCAIP) – Specification*
SIS-TR 91101:2014	*Digital social alarm – Social care alarm internet protocol (SCAIP) – Test specification*
SIS-TR 91102:2014	*Digital social alarm – Social care alarm internet protocol (SCAIP) – Implementation guideline*

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1 Scope

This standard specifies the Social Care Alarm Internet Protocol, SCAIP. This standard describes how the protocol handles a multimedia communication stream and how to send event messages between the alarm sender and the alarm receiver over an IP communication network such as Internet.

The data exchange of the protocol is defined as an XML schema including the alarm types, codes and additional information required to fulfil the requested functionality. The protocol uses a widely used and standardized application layer protocol, SIP (Session Initiation Protocol), for creating, modifying and terminating the session.

2 Normative references

This clause specifies the following documents as indispensable for the use of this document. For dated references, only the dated reference applies. For undated references the latest edition of the document applies (including all amendments).

RFC 2617, *HTTP Authentication: Basic and Digest Access Authentication*

RFC 3261, *SIP: Session Initiation Protocol*

RFC 3264, *An Offer/Answer model with Session Description Protocol (SDP)*

RFC 3428, *SIP Extension for Instant Messaging*

RFC 3550, *RTP: A Transport Protocol for Real-time Applications*

RFC 4733, *RTP Payload for DTMF Digits, Telephony Tones and Telephony Signals*

3 Terms and Abbreviations

For the use of this document the following terms and abbreviations apply.

3.1
alarm receiver
alarm receiving centre
system part which provides facilities for communication with a number of controllers, and providing the alarm receiving and information processing system as an interface to the alarm recipient

3.2
codec
device capable of encoding and decoding a digital data stream or signal

3.3
controller
alarm sender
interface between one or more Local Units and the alarm transmission system or alarm recipient

3.4
heartbeat
periodic event generated by hardware or software to indicate normal operation or to synchronise parts of a system

3.5
in-band signalling
sending of control information within the same band or channel used for voice

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3.6
interconnections
transmission system that provides the communication between trigger devices and local unit and controller

3.7
Local Unit
interface between the user and the controller which enables two-way speech

3.8
multimedia
media and content that use a combination of different content forms as text, audio, still images, animation, video or interactivity

3.9
polling
sampling of a device to synchronise an activity

3.10
protocol
system of digital rules for message exchange within or between computers

3.11
social alarm system
telecare system
system providing 24 h facilities for alarm triggering, identification, signal transmission, alarm reception, two-way speech communication, reassurance and assistance, for use by persons considered to be at risk

Abbreviation	Definition
ARC	Alarm Receiving Centre
ASCII	American Standard Code for Information Interchange
GGA	Global Positioning System Fix Data
GPS	Global Positioning System
GSM	Global System for Mobile Communications
IP	Internet Protocol
ISO	International Organization for Standardization
LUC	Local Unit and Controller
NMEA	National Marine Electronics Association
POTS	Plain Old Telephone Service
RFC	Request for Comments
RTP	Real-time Transport Protocol
SRTP	Secure Real-time Transport Protocol
SDP	Session Description Protocol
SIP	Session Initiation Protocol
SIP-PP	Session Initiation Protocol, Peer-to-peer
TLS	Transport Layer Security
UCS	Universal Character Set

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Abbreviation	Definition
UDP	User Datagram Protocol
UTF	UCS Transformation Formats
XML	eXtensible Markup Language
URI	Uniform Resource Identifier

4 Use Case 1: Event without voice- or multimedia communication

4.1 General

Figure 1 shows a communication model with an alarm sender communicating over the network, e.g. Internet, with SCAIP as protocol and an alarm receiver as endpoint. Messages are initiated from either an end user, e.g. human, or a device and could have the function either as an alarm or a status message.

Figure 1 — Communication setup

Figure 2 shows a message exchange between the alarm sender and the alarm receiver.

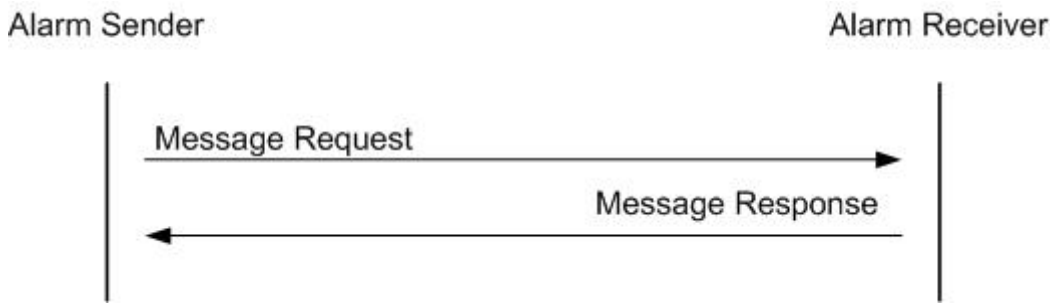

Figure 2 — Message exchange

In this use case, the alarm sender sends a Message Request to the alarm receiver. Each Message Request shall be acknowledged by the alarm receiver using a Message Response.

An event shall be considered successfully handled when the status number (status-number, see table 2) of the Message Response is set to 0.

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4.2 Event not treated by alarm receiver

If the alarm receiver cannot determine the event as treated it shall return a Message Response with status message (status-number, see table 2) set to 4. This shall cause the alarm sender to resend the same Message Request after 5 to 20 seconds.

This polling sequence shall continue for at least 3 minutes but shall not continue for more than 30 minutes.

4.3 Event information update

If the alarm sender has new information related to an event that is being processed the alarm sender shall send a Message Request with message type (message-type, see Table 1) set to Information update and all mandatory data elements set to the same value as the initial Message Request.

4.4 Aborting message session

The alarm sender shall stop resending the Message Request and continue with an alternative action if the alarm sender has not received a Message Response with status number (status-number, see table 2) set to 0 within a specified time. In such case it shall abort the alarm by sending a Message Request with message type (message-type, see table 1) set to Reset and reference (reference, see table 1) set to the same value as in the initial Message Request.

4.5 Heartbeat

The alarm sender shall poll the alarm receiver periodically with a Message Request having message type (message-type, see table 1) set to Heartbeat. The alarm receiver shall return a Message Response with status number (status-number, see table 2) set to 0.

The heartbeat period should be minimum 1 minute and maximum 1440 minutes.

5 Use case 2: Event with voice or multimedia communication

In this use case the alarm sender will send a Message Request as well as establish a voice or multimedia session. The message exchange shall be handled the same way as in use case 1, shown in figure 1, with the exception that the voice or multimedia sessions should only be initiated by the alarm sender after it has received a Message Response with status number (status-number, see table 2) set to 0 and media reply (media-reply, see table 2) set to 1 or more. Upon reception of the Message Response, the alarm sender should initiate the voice session within 60 seconds.

NOTE The alarm receiver of the Message Request may or may not be the same as the alarm receiver of the voice or multimedia communication. The alarm sender may be able to connect to multiple alarm receivers for different events.

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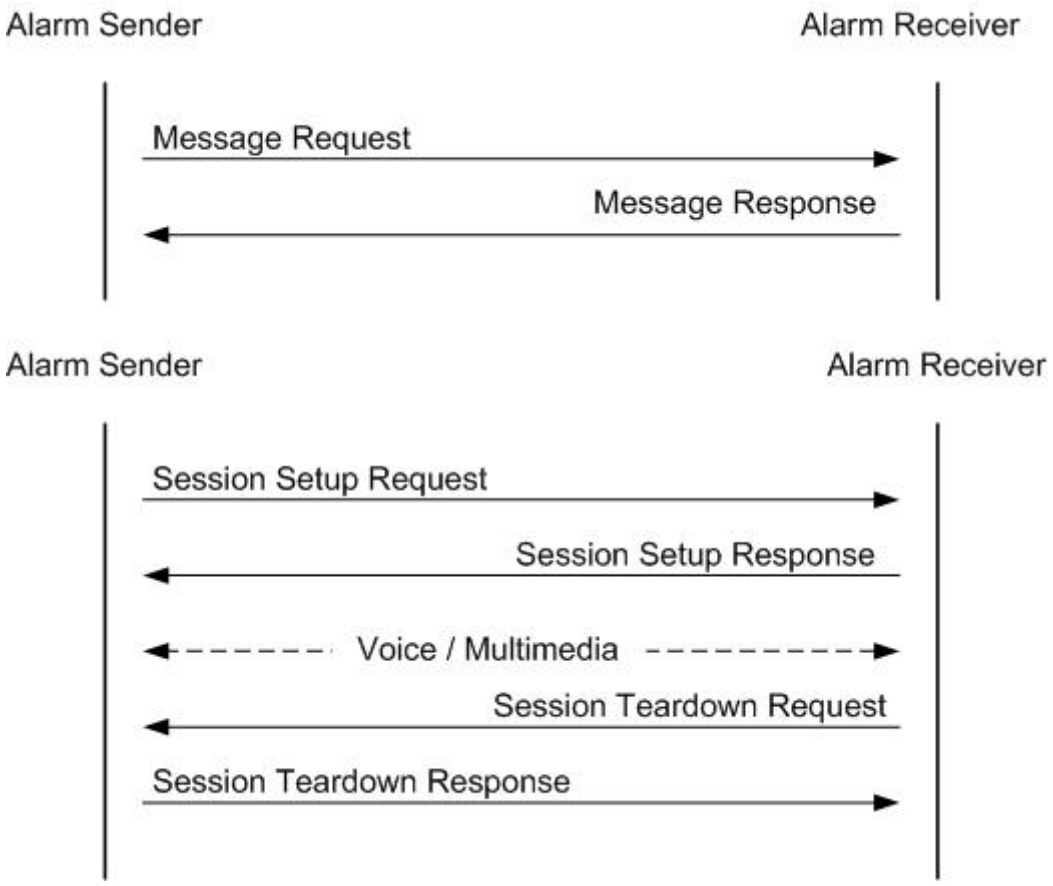

Figure 3 — Message exchange with voice or multimedia communication

Figure 3 shows a message exchange combined with the establishment of a voice or multimedia communication.

The voice or multimedia communication shall be initiated by the alarm sender. Once the communication is established it may be modified or terminated by either the alarm sender or the alarm receiver.

Some functions should be controlled by single DTMF codes from the alarm receiver to enable a basic control facility from a handset with a traditional telephony keyboard.

6 Message format description

6.1 General

SCAIP uses an XML format for the messages to represent the alarms and information codes. The messages shall conform to a specific XML schema that should be used to verify the messages. The XML schema is defined in Annex F.

Table 1 and Table 2 provides information about the data elements. The column “Length in characters” provides the minimum and maximum number of characters of the text in a present data element. An asterisk (*) indicates that the data element is mandatory. If a non-mandatory data element is excluded the receiver of the Message Request or Message Response should treat it as defined for “omitted”.

EXAMPLE 1 0/2-4 without an asterisk (*) means that the data element may be omitted. When it is omitted it shall have a length of 0 characters when it is present the length shall be 2, 3 or 4 characters.

EXAMPLE 2 1-16 * means that data element is mandatory and shall have a length of 1 to 16 characters.

Content values enclosed between apostrophes are to be interpreted as the text string between them and shall be coded in US-ASCII format if not otherwise stated.

EXAMPLE 3 “1” means the US-ASCII value of the character 1.

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6.2 Message Request

Message Requests shall be encoded as a list of child XML tags and data elements contained in the root tag message-request according to the template below. The template defines the XML encoding of all possible data elements. The data elements are specified in table 1.

```
<mrq>
     <ref>reference</ref>
     <ver>version</ver>
     <sco>system-config</sco>
     <cha>call-handling</cha>
     <mty>message-type</mty>
     <hbo>heartbeat-options</hbo>
     <cid>controller-id</cid>
     <dty>device-type</dty>
     <did>device-id</did>
     <dco>device-component</dco>
     <dte>device-text</dte>
     <crd>caller-id</crd>
     <stc>status-code</stc>
     <stt>status-text</stt>
     <pri>priority</pri>
     <lco>location-code</lco>
     <lva>location-value</lva>
     <lge>
          <geo>wgs-pos</geo>
          <tim>time-stamp</tim>
          <gga>gga-pos</gga>
     </lge>
     <lte>location-text</lte>
     <ico>info-code</ico>
     <ite>info-text</ite>
     <ame>additional-message</ame>
</mrq>
```

Table 1 – XML description for Message Requests

Data element name	XML tag	Description of content	Length in characters
message-request	<mrq>	Compound element for the Message Request	
reference	<ref>	The reference is used to match the Message Response against the Message Request. The reference number shall differ between subsequent Message Sessions. The valid characters are: “A” through “Z”, “a” through “z” and “0” through “9”.	1-16 *
version	<ver>	Protocol version written as two digits, a period and two digits. EXAMPLE “01.23” When omitted, “01.00” shall be assumed. The valid characters are “0” through “9”	0/5
system-config	<sco>	System configuration and information Omitted or “0” = Local unit and controller “1” = Grouped equipment with supervisor off duty “2” = Grouped equipment with supervisor on duty “3” = Grouped equipment with supervisor on duty acting as an alarm receiver	0/1

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Data element name	XML tag	Description of content	Length in characters
`call-handling`	`<cha>`	Omitted or “0” = Normal outgoing call from the alarm sender “1” = Callback requested	0/1
`message-type`	`<mty>`	Omitted or “ME” = Message “RE” = Reset “IN” = Information update within the Message Session “PI” = Heartbeat	0/2
`heartbeat-options`	`<hbo>`	Omitted or “0” = Alarm sender can not handle heartbeat time adjustments from the alarm receiver. “001” = Alarm sender can handle heartbeat time adjustments from the alarm receiver. Other = Unspecified for future use.	0/1-3
`controller-id`	`<cid>`	ID or account number for the controller that is calling. The valid characters for this field are “0” through “9”.	1-32 *
`device-type`	`<dty>`	The device type contains the specification of the device. Valid device types are defined in Annex A.	1-4 *
`device-id`	`<did>`	The device ID is used to indicate the identity of a physical device causing the event. The valid characters for this field are characters with US-ASCII encoding 32 through 126.	0/1-16
`device-component`	`<dco>`	The device component is used to further indicate the source of an event within a device. Valid device component codes are defined in Annex B	0/1-3
`device-text`	`<dte>`	Device text description. The text is only aimed to give additional information. Not to contradict the device type or component. The valid character set for this field is Unicode and UTF-8 encoding shall be used	0/1-32
`caller-id`	`<crd>`	Media and telephone number or SIP URI of the alarm sender. EXAMPLE 1 “gsm:+46701445566” One XML element shall be used for each available media. Valid media types are: “no-voice:” “gsm:” “sip:” “sip-pp:” If the telephone number or SIP URI is unknown but can be used for outgoing call the number should be omitted. Omitted = “sip:” EXAMPLE 2 “gsm:”	0/1-256
`status-code`	`<stc>`	The status code is used to indicate the device state. Valid status codes are defined in Annex C When omitted, alarm state shall be assumed.	0/1-4
`status-text`	`<stt>`	Status text description. The text shall only give additional information and shall not contradict with the status code. The valid character set for this field is Unicode and UTF-8 encoding shall be used	0/1-32

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Data element name	XML tag	Description of content	Length in characters
priority	<pri>	A single-digit priority field, to indicate that a priority over other messages has been assigned to this event. “0” = lowest priority “1”-”8“= increasing priority “9“ = highest priority Omitted = “0”	0/1
location-code	<lco>	Location information to allow detailed locations of events to be indicated. The location code can also refer to a location value when needed. Valid location codes are defined in Annex D	0/1-3
location-value	<lva>	Further specification of the location-code. It can be either a simple numbering of a fixed location, an area or a section number. The valid characters for this field are characters with US-ASCII encoding 32 through 126.	0/1-2
location-geo	<lge>	When a geographical coordinate or time stamp is added the additional sub elements below shall be used.	
wgs-pos	<geo>	2D microformat of coordinate reference system WGS-84 expressed as latitude and longitude in degrees, extracted from “geo” specified in RFC 5870 [16]. EXAMPLE “59.336111,18.072778”	0/1-23
time-stamp	<tim>	Additional time stamp according to ISO 8601 [17] with time zone expressed as difference to UTC in hours. Timestamp in Central European Time (CET): YYYY-MM-DDThh:mm:ss+01	0/22
gga-pos	<gga>	GPS, NMEA sentence GGA - essential fix data which provide 3D location and accuracy data $GPGGA,hhmmss.ss,llll.ll,a,yyyyy.yy, a,x,xx,x.x,x.x,M,x.x,M,x.x,xxxx*hh	0/81
location-text	<lte>	Text description of the location from where the Message Request is raised. EXAMPLE “Chapel” The text is only aimed to give additional information. Not to contradict the location code or value. The valid character set for this field is Unicode and UTF-8 encoding shall be used	0/1-32
info-code	<ico>	Additional coded information about the event. Valid info codes are defined in Annex E	0/1-3
info-text	<ite>	Additional text information about the event The text is only aimed to give additional information. Not to contradict the info code. The valid character set for this field is Unicode and UTF-8 encoding shall be used	0/1-128

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Data element name	XML tag	Description of content	Length in characters
additional-message	<ame>	Additional message shall be used if total length of the message request exceeds 1300 bytes. If the Message Request exceeds the limit it shall be sent as several Message Requests marked with number of additional message after present message EXAMPLE if a Message Request is 1700 bytes it shall be divided in two messages whereof additional messages is stated to 1 in the first message and 0 in the second (last) message. Omitted or “0”= Last message “n” = Number of additional messages after present message The valid characters for this field are “0” through “9”	0/1-4

6.3 Message Response

Message Responses shall be encoded as a list of child XML tags and data elements contained in the root tag message-response according to the template below. The template defines the XML encoding of all possible data elements. The data elements are specified in table 2.

```
<mrs>
     <ref>reference</ref>
     <snu>status-number</snu>
     <ste>status-text</ste>
     <cve>common-version</cve>
     <mre>media-reply</mre>
     <cre>callhandling-reply</cre>
     <tnu>transferred-number</tnu>
     <hbi>heartbeat-interval</hbi>
</mrs>
```

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Table 2 – XML description for Message response

Data element name	XML Tag	Description of content	Length in characters
`message-response`	`<mrs>`	Compound element for the Message Response	
`reference`	`<ref>`	The reference number extracted from <mrq>.	1-16 *
`status-number`	`<snu>`	**Alarm processing related status-numbers:** “0” = Ok. The alarm has been handled successfully. No further actions is necessary “4” = Hold. The alarm is being processed. Resend alarm for updated status information. “5” = Not treated or not distributed. The alarm couldn´t be handled. Try next alarm action “6” = Busy. The alarm couldn´t be handled right now. Try again later or proceed with next alarm action. **Message related status-numbers:** “1” = message to long “2” = invalid format “3” = wrong data content “7” = mandatory tag is missing “10” = reference missing “99” = undefined error	1-3 *
`status-text`	`<ste>`	Text information about the status number. The text is only aimed to give additional information. Not to contradict the status number. The valid characters for this field are characters with US-ASCII encoding 32 through 126.	0-128
`common-version`	`<cve>`	Protocol version to use that shall be equal or lower than the version in the corresponding Message Request. Omitted = Initial version, “01.00” The valid characters are “0” through “9”.	0/5
`media-reply`	`<mre>`	Omitted or “0” = no voice call “1” = duplex voice call “2” = simplex voice call, microphone (LUC speaker muted) “3” = simplex voice call, speaker (LUC microphone muted)	0/1-2
`callhandling-reply`	`<cre>`	Reply to Message Request *call-handling*, <cha>. “01”-”60” (nn) = Callback accepted, open auto-answer channel for nn minutes. Requires that the alarm receiver have registered the valid alarm sender URI/Telephone number or a number or identity contained in the *caller-id*, <crd> Omitted or “61” = set-up a voice call to a pre-defined receiver “62” = set-up a voice call to transferred URI/Telephone number defined in *transferred-number*, <tnu>	0/2
`transferred-number`	`<tnu>`	Media and telephone number or SIP URI transferred for *call-handling-reply*, <cre>. EXAMPLE: “gsm:+46701445566” One XML element shall be used for each available media. Valid media types are: “gsm:” “sip:” “sip-pp:”	0/1-256

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Data element name	XML Tag	Description of content	Length in characters
`heartbeat-interval`	`<hbi>`	Heartbeat interval The alarm receiver can adjust the alarm sender heartbeat interval if the alarm sender has indicated it can handle such adjustment by setting heartbeat options<hbo> to “001”. “1”-”1440” = Time in minutes between consecutive heart-beats. Omitted = Default value specified by the alarm sender shall be used.	0/1-4

7 DTMF code

The audio functions should be controlled by single DTMF codes from the alarm receiver. These codes are only applicable for voice or multimedia communication. The duration of the DTMF codes should be between 100 ms and 500 ms.

EXAMPLE 1 DTMF codes can be sent as SIP INFO, RFC 4733 or as in-band audio tones.

Table 3 – DTMF codes

DTMF code	Action
0	Clear Down. (voice call received correctly by the alarm receiver)
1	Transmitter speaker volume down (-)
2	
3	Transmitter speaker volume up (+)
4	Duplex
5	
6	
7	Simplex voice call, microphone (LUC speaker muted)
8	Simplex voice call, speaker (LUC microphone muted)
9	Output control
*	
#	

8 Authentication

The application layer protocol shall support authentication on a per connection basis, at least the same level of encryption and authentication mechanisms as SIP. Authentication shall be either user specific, where each alarm sender logs in on a separate account or group specific, where all alarm senders log in on a common account.

The authentication mechanism shall support HTTP Authentication [9]. HTTP Basic authentication scheme shall not be used.

NOTE Since SCAIP by itself is not protected by any encryption or authentication mechanism it is important that the underlying application layer provide security and message integrity. The application layer protocol defined by this document to carry the messages, SIP, supports security mechanism such as TLS [10].

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Alarm receivers shall only accept and process messages from authenticated, and thus authorised, alarm senders.

9 Sessions

9.1 Message

All XML content shall be carried by SIP MESSAGE. An example of a SIP MESSAGE with XML content is shown below

EXAMPLE A Message Request carried by SIP MESSAGE [4] is shown below. Note that the "*Content-Type*" is set to "application/scaip+xml". This value shall be used when carrying a message. The body of the SIP MESSAGE holds the message in XML syntax as defined by the schema in Annex F.

```
MESSAGE sip:user2@domain.com SIP/2.0
Via: SIP/2.0/UDP user1.domain.com;branch=z9hG4bK776sgdkse
To: sip:user2@domain.com
From: sip:user1@domain.com;tag=49583
Call-ID: asd88asd77a@1.2.3.4
CSeq: 1 MESSAGE
Max-Forwards: 70
Content-Type: application/scaip+xml
Content-Length: 144

<mrq>
  <ref>1234</ref>
  <cid>123456</cid>
  <dty>0001</dty>
  <stc>001</stc>
</mrq>
```

9.2 Voice or multimedia

When using VoIP, initiation, modification and termination of the voice or multimedia communication shall be done using methods defined by SIP [2].

NOTE During the establishment of the voice or multimedia communication, media capabilities and codecs as well as IP addresses and ports to be used are negotiated. The multimedia content is usually packetized and transported using RTP [5].

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Annex A
(normative)

Codes for device types

Table A.1 – Device types

Code	Meaning	Note
0000	Unknown	
0001	Undefined	
0002	Local unit & controller	
0003	Personal trigger/pendent	
0004	Fixed trigger	
0005	Fall trigger/detector	
0006	Personal attack pendent	
0007	Panic alarm button	
0008	Duty switch remote	
0009	Duty switch local	
0010	Activity detector	Also for inverted use as inactivity detection.
0011	Pill dispenser	
0012	Bed monitor	
0013	Mat sensor	
0014	Door sensor	
0015	Wandering sensor	Normally used together with location.
0016	Enuresis detector	
0017	Epilepsy detector	
0018	Occupancy detector	
0019	Heart rate monitor	
0020	Environmental monitor	
0021	(Natural) gas detector	
0022	Lighting circuit monitor	
0023	Smoke detector	
0024	Heating system monitor	
0025	Heat detector	
0026	Carbon monoxide gas detector	

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Code	Meaning	Note
0027	Bogus caller switch	
0028	Intruder detector	
0029	Boundary detector	
0030	Automatic) man down detector	
0031	System monitor	
0032	Communication link monitor	
0033	Flood detector	
0034	Bath sensor	
0035	Latch key	
0036	Perimeter	
0037	Sprinkler	
0038	Tamper	
0039	Printer	
0040	Router	
0041	Power supply, DC	
0042	Electric lock	
0043	Access controller	
0044	Chair monitor	
0045	Controller	
0046	Elevator alarm unit	
0047	Fire alarm system	
0048	Local unit	
0049	Monitoring device	Undefined
0050	Radio unit	Undefined
0051	Tracking device	
0052-0899	Allocated for future use	
0900-0999	Reserved for flexible use without notice	The device-codes within this interval are only valid together with one or a specified set of unique control-ler-id
1000-9999	Unspecified for future use	

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Annex B
(normative)

Codes for device components

Table B.1 – Device components

Code	Meaning	Note
000	Unknown	
001	Undefined	
002	Button 1	
003	Button 2	
004	Button 3	
005	Button 4	
006	Button 5	
007	Switch 1	
008	Switch 2	
009	Switch 3	
010	Switch 4	
011	Switch 5	
012	Output 1	
013	Output 2	
014	Output 3	
015	Output 4	
016	Output 5	
017	Automatic circuit 1	
018	Automatic circuit 2	
019	Automatic circuit 3	
020	Automatic circuit 4	
021	Automatic circuit 5	
022	Ethernet circuit	
023	Memory circuit	
024	Microcontroller	
025	Module 1	
026	Module 2	
027	Module 3	
028	Module 4	
029	Module 5	
030-089	Allocated for future use	
090-099	Reserved for flexible use without notice	The device-components within this interval are only valid together with one or a specified set of unique controller-id's
100-999	Unspecified for future use	

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Annex C
(normative)

Status codes

Table C.1 – Status codes

Code	Meaning	Note
0000	Unknown	
0001	Undefined	
0002	Abort	
0003	Absent/away	
0004	Accepted	
0005	Acknowledge	
0006	Activated	
0007	Active	
0008	Added	
0009	Alarm (automatic)	
0010	Alarm (manual)	
0011	Alert	
0012	Armed	
0013	Assigned	
0014	Battery failure	
0015	Battery high/full/recharged	
0016	Battery low	
0017	Busy	
0018	Bypass	
0019	Cancel	
0020	Changed	
0021	Closed	
0022	Closed to long	
0023	Communication link, IP, failure	
0024	Communication link, mobile network, failure	
0025	Communication link, POTS, failure	
0026	Communication link, primary, failure	
0027	Communication link, redundant, failure	
0028	Deleted	
0029	Denied	
0030	Disarmed	
0031	Disconnected	
0032	Error, code 1	
0033	Error, code 2	

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Code	Meaning	Note
0034	Error, code 3	
0035	Error, code 4	
0036	Error, code 5	
0037	Error, code 6	
0038	Error, code 7	
0039	Error, code 8	
0040	Error, code 9	
0041	Executed	
0042	Failure	
0043	Frequency drifted	
0044	Frequency drifted to lower level	
0045	Frequency drifted to upper level	
0046	High level	
0047	In progress	
0048	In Service (Fault Rectified)	
0049	Inactive	
0050	Initiated	
0051	Interrupted	
0052	Jammed	
0053	Level 1	
0054	Level 2	
0055	Level 3	
0056	Level 4	
0057	Level 5	
0058	Level 6	
0059	Level 7	
0060	Level 8	
0061	Level 9	
0062	Log	
0063	Loss	
0064	Low level	
0065	Maintenance, scheduled, needed	
0066	Maintenance, urgently needed	
0067	Malfunction	
0068	Memory full	
0069	Non-active	
0070	Normal state	
0071	Occupied	
0072	Off (indication)	
0073	Offline	
0074	On (indication)	

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Code	Meaning	Note
0075	Online	
0076	Open	
0077	Open to long	
0078	Out of Service	
0079	Overflow	
0080	Pending	
0081	Pill not taken	
0082	Pill reminder	
0083	Pill taken	
0084	Planned	
0085	Power failure (mains)	
0086	Power restored (mains)	
0087	Presence, planned	
0088	Presence, upon alarm	
0089	Privacy Switch Operated	
0090	Pulse	
0091	Re-established	
0092	Reset (automatic)	
0093	Reset (manual)	
0094	Restoral	
0095	RF Interference	
0096	Started	
0097	Stopped/ended	
0098	Substituted	
0099	Succeeded	
0100	Temperature high	
0101	Temperature low	
0102	Test transmission, primary channel	
0103	Test transmission, redundant channel	
0104	Time out	
0105	Trouble	
0106	Trespass	
0107	Unauthorized access	
0108	Unbypass	
0109	Verified	
0110	Assistance	
0111	Check-out (intervention completed)	
0112	Communication link, IP, re-established	
0113	Communication link, mobile network, re-established	

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Code	Meaning	Note
0114	Communication link, POTS, re-established	
0115	Communication link, primary, re-established	
0116	Communication link, redundant, re-established	
0117	Lying in bed	
0118	Locked	
0119	Movement	
0120	Occupancy	
0121	Present, home	
0122	Radio interference	
0123	Radio test transmission missing	
0124	Radio test transmission re-established	
0125	Tamper alarm	
0126	Temperature rate of rise	
0127	Unlocked	
0128-0899	Allocated for future use	
0900-0999	Reserved for flexible use without notice	The status-codes within this interval are only valid together with one or a specified set of unique controller-id's
1000-9999	Unspecified for future use	

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Annex D
(normative)

Location codes

Table D.1 – Location codes

Code	Meaning	Note
000	Unknown	
001	Undefined	
002	Home	
003	Away	
004	Indoors	
005	Outdoors	
006	Inside	
007	Outside	
008	Area	
009	Zone	
010	Basement	
011	Bath room / WC	
012	Bed room	
013	Dining room	
014	Floor	
015	Garage	
016	Garden	
017	Garden, back	
018	Garden, front	
019	Geo fence	
020	Gym	
021	Kitchen	
022	Laundry	
023	Relax	
024	Sauna	
025	Stairs	
026	Study room	
027	Work shop	
028	Loft	
029	Shed	
030	Temporary address	Registered at ARC. Several addresses (2-99) require valid location value <lva>.
031	Summer house	
032	Winter cottage	
033	Mobile w/o fixed location	

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Code	Meaning	Note
034	Geographic position attached	Requires valid location value <geo>.
035	Geographic GPS position attached	Requires valid location value <gga>.
036	Property / estate	
037	HWC, Handicap WC	
038	Door	
039	Front door	
040	Back door	
041	Fire door	
042	Hallway	
043	Living room	
044	Apartment	
045	Bungalow	
046	Lounge	
047	Communal area	
048	Office	
049	Room	
050	Site	
051	Ward	
052	Wing	
053-089	Allocated for future use	
090-099	Reserved for flexible use without notice	The location-codes within this interval are only valid together with one or a specified set of unique controller-id's.
100-999	Unspecified for future use	

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Annex E
(normative)

Info codes

Table E.1 – Info codes

Code	Meaning	Note
000-089	Allocated for future use	
090-099	Reserved for flexible use without notice	The info-codes within this interval is only valid together with one or a specified set of unique controller-id's
100-999	Unspecified for future use	

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Annex F
(informative)

XML schema

An example of the XML schema is listed below and the latest version is found at: http://www.sis.se/tk574

```
<?xml version="1.0" encoding="UTF-8" ?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema" target-
Namespace="http://www.hi.se/2012/XScaipXML" elementFormDefault="qualified">
  <xs:element name="mrq">
    <xs:complexType>
      <xs:all>
        <xs:element name="ref">
          <xs:simpleType>
            <xs:restriction base="xs:string">
              <xs:pattern value="[0-9a-zA-Z]{1,16}" />
            </xs:restriction>
          </xs:simpleType>
        </xs:element>
        <xs:element name="ver" minOccurs="0">
          <xs:simpleType>
            <xs:restriction base="xs:string">
              <xs:pattern value="[0-9]{2}\.[0-9]{2}" />
            </xs:restriction>
          </xs:simpleType>
        </xs:element>
        <xs:element name="sco" minOccurs="0">
          <xs:simpleType>
            <xs:restriction base="xs:string">
              <xs:pattern value="[1-3]{1}" />
            </xs:restriction>
          </xs:simpleType>
        </xs:element>
        <xs:element name="cha" minOccurs="0">
          <xs:simpleType>
            <xs:restriction base="xs:string">
              <xs:pattern value="[0-9]{1}" />
            </xs:restriction>
          </xs:simpleType>
        </xs:element>
        <xs:element name="mty" minOccurs="0">
          <xs:simpleType>
            <xs:restriction base="xs:string">
              <xs:pattern value="(ME)|(RE)|(IN)|(PI)" />
            </xs:restriction>
          </xs:simpleType>
        </xs:element>
        <xs:element name="hbo" minOccurs="0">
          <xs:simpleType>
            <xs:restriction base="xs:string">
              <xs:pattern value="[0-3]{1}" />
            </xs:restriction>
          </xs:simpleType>
        </xs:element>

        <xs:element name="cid">
          <xs:simpleType>
            <xs:restriction base="xs:string">
```

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```
      <xs:pattern value="[0-9]{1,16}" />
    </xs:restriction>
  </xs:simpleType>
</xs:element>
<xs:element name="dty">
  <xs:simpleType>
    <xs:restriction base="xs:string">
      <xs:pattern value="[0-9]{1,4}" />
    </xs:restriction>
  </xs:simpleType>
</xs:element>
<xs:element name="did" minOccurs="0">
  <xs:simpleType>
    <xs:restriction base="xs:string">
      <xs:minLength value="1" />
      <xs:maxLength value="8" />
    </xs:restriction>
  </xs:simpleType>
</xs:element>
<xs:element name="dco" minOccurs="0">
  <xs:simpleType>
    <xs:restriction base="xs:string">
      <xs:pattern value="[0-9]{1,3}" />
    </xs:restriction>
  </xs:simpleType>
</xs:element>
<xs:element name="dte" minOccurs="0">
  <xs:simpleType>
    <xs:restriction base="xs:string">
      <xs:minLength value="1" />
      <xs:maxLength value="32" />
    </xs:restriction>
  </xs:simpleType>
</xs:element>
<xs:element name="crd" minOccurs="0">
  <xs:simpleType>
    <xs:restriction base="xs:string">
      <xs:minLength value="1" />
      <xs:maxLength value="256" />
    </xs:restriction>
  </xs:simpleType>
</xs:element>
<xs:element name="stc" minOccurs="0">
  <xs:simpleType>
    <xs:restriction base="xs:string">
      <xs:pattern value="[0-9]{1,4}" />
    </xs:restriction>
  </xs:simpleType>
</xs:element>
<xs:element name="stt" minOccurs="0">
  <xs:simpleType>
    <xs:restriction base="xs:string">
      <xs:minLength value="1" />
      <xs:maxLength value="32" />
    </xs:restriction>
  </xs:simpleType>
</xs:element>
<xs:element name="pri" minOccurs="0">
  <xs:simpleType>
    <xs:restriction base="xs:string">
      <xs:pattern value="[0-9]{1}" />
    </xs:restriction>
```

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```
      </xs:simpleType>
    </xs:element>
    <xs:element name="lco" minOccurs="0">
      <xs:simpleType>
        <xs:restriction base="xs:string">
          <xs:pattern value="[0-9]{1,3}" />
        </xs:restriction>
      </xs:simpleType>
    </xs:element>
    <xs:element name="lva" minOccurs="0">
      <xs:simpleType>
        <xs:restriction base="xs:string">
          <xs:minLength value="1" />
          <xs:maxLength value="2" />
        </xs:restriction>
      </xs:simpleType>
    </xs:element>
    <xs:element name="lge" minOccurs="0">
      <xs:complexType mixed="true">
        <xs:all>
          <xs:element name="geo" minOccurs="0">
            <xs:simpleType>
              <xs:restriction base="xs:string">
                <xs:pattern value="[0-9\.,]{1,23}" />
              </xs:restriction>
            </xs:simpleType>
          </xs:element>
          <xs:element name="tim" minOccurs="0">
            <xs:simpleType>
              <xs:restriction base="xs:string">
                <xs:pattern value="[0-9\-:\+T]{22}" />
              </xs:restriction>
            </xs:simpleType>
          </xs:element>
          <xs:element name="gga" minOccurs="0">
            <xs:simpleType>
              <xs:restriction base="xs:string">
                <xs:minLength value="0" />
                <xs:maxLength value="81" />
              </xs:restriction>
            </xs:simpleType>
          </xs:element>
        </xs:all>
      </xs:complexType>
    </xs:element>
    <xs:element name="lte" minOccurs="0">
      <xs:simpleType>
        <xs:restriction base="xs:string">
          <xs:minLength value="1" />
          <xs:maxLength value="32" />
        </xs:restriction>
      </xs:simpleType>
    </xs:element>
    <xs:element name="ico" minOccurs="0">
      <xs:simpleType>
        <xs:restriction base="xs:string">
          <xs:pattern value="[0-9]{1,3}" />
        </xs:restriction>
      </xs:simpleType>
    </xs:element>
    <xs:element name="ite" minOccurs="0">
      <xs:simpleType>
```

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```
          <xs:restriction base="xs:string">
            <xs:minLength value="1" />
            <xs:maxLength value="128" />
          </xs:restriction>
        </xs:simpleType>
      </xs:element>
      <xs:element name="ame" minOccurs="0">
        <xs:simpleType>
          <xs:restriction base="xs:string">
            <xs:pattern value="[01]?" />
          </xs:restriction>
        </xs:simpleType>
      </xs:element>
    </xs:all>
  </xs:complexType>
</xs:element>
<xs:element name="mrs">
  <xs:complexType>
    <xs:all>
      <xs:element name="ref">
        <xs:simpleType>
          <xs:restriction base="xs:string">
            <xs:pattern value="[0-9a-zA-Z]{1,16}" />
          </xs:restriction>
        </xs:simpleType>
      </xs:element>
      <xs:element name="snu">
        <xs:simpleType>
          <xs:restriction base="xs:string">
            <xs:pattern value="[0-9]{1,5}" />
          </xs:restriction>
        </xs:simpleType>
      </xs:element>
      <xs:element name="ste" minOccurs="0">
        <xs:simpleType>
          <xs:restriction base="xs:string">
            <xs:minLength value="1" />
            <xs:maxLength value="128" />
          </xs:restriction>
        </xs:simpleType>
      </xs:element>
      <xs:element name="cve" minOccurs="0">
        <xs:simpleType>
          <xs:restriction base="xs:string">
            <xs:pattern value="[0-9]{2}\.[0-9]{2}" />
          </xs:restriction>
        </xs:simpleType>
      </xs:element>
      <xs:element name="mre" minOccurs="0">
        <xs:simpleType>
          <xs:restriction base="xs:string">
            <xs:pattern value="[0-9]{2}" />
          </xs:restriction>
        </xs:simpleType>
      </xs:element>
      <xs:element name="cre" minOccurs="0">
        <xs:simpleType>
          <xs:restriction base="xs:string">
            <xs:pattern value="[0-9]{2}" />
          </xs:restriction>
        </xs:simpleType>
      </xs:element>
```

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```
        <xs:element name="tnu" minOccurs="0">
          <xs:simpleType>
            <xs:restriction base="xs:string">
              <xs:minLength value="1" />
              <xs:maxLength value="256" />
           </xs:restriction>
          </xs:simpleType>
        </xs:element>
        <xs:element name="hbi" minOccurs="0">
          <xs:simpleType>
            <xs:restriction base="xs:string">
              <xs:pattern value="[0-4]{1}" />
            </xs:restriction>
          </xs:simpleType>
        </xs:element>
      </xs:all>
    </xs:complexType>
  </xs:element>
</xs:schema>
```

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Bibliography

[1] RFC 2119, *Key words to use for RFCs to indicate requirement level*

[2] RFC 3261, *SIP: Session Initiation Protocol*

[3] RFC 6455, *The Websocket Protocol*

[4] RFC 3428, *SIP Extension for Instant Messaging*

[5] RFC 3550, *RTP: Transport Protocol for Real-time Applications*

[6] RFC 3711, *SRTP: The Secure Real-time Transport Protocol*

[7] RFC 4733, *RTP Payload for DTMF Digits, Telephony Tones and Telephony Signals*

[8] RFC 6086, *SIP INFO Method and Package Framework*

[9] RFC 2617, *HTTP Authentication: Basic and Digest Access Authentication*

[10] RFC 5246, *TLS: The Transport Layer Security Protocol*

[11] RFC 3263, *Locating SIP Servers*

[12] RFC 3264, *An Offer/Answer model with Session*

[13] RFC 4568, *SDP Security Descriptions for Media Streams*

[14] RFC 3830, *MIKEY: Multimedia Internet KEYing*

[15] RFC 5764, *DTLS Extension to Establish Keys for SRTP*

[16] RFC 5870, *A Uniform Resource Identifier for Geographic Locations (“geo” URI)*

[17] ISO 8601, *Representation of dates and times*

[18] G.711 (11/88), *Pulse code modulation (PCM) of voice frequencies*

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Anteckningar/Notes

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Anteckningar/Notes

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Ordlista

Här har vi samlat de förkortningar som oftast används i standardiseringssammanhang och förklarat dem kortfattat. Förkortningarna är sorterade i alfabetisk ordning.

CEN	European Committee for Standardization (Comité Européen de Normalisation). Utarbetar Europastandarder för områden som inte täcks av CENELEC.
CENELEC	European Committee for Electrotechnical Standardization (Comité Européen de Normalisation Electro-technique). Utarbetar Europastandarder inom el.
CWA	CEN/CENELEC Workshop Agreement. Tekniskt dokument utarbetat av CEN- och CENELEC-organiserad arbetsgrupp.
EN	Europastandard från CEN/CENELEC.
ETSI	European Telecommunications Standards Institute. De utarbetar Europastandarder inom telekommunikationsområdet.
FDIS	Final Draft International Standard. Slutligt förslag till global standard från IEC eller ISO.
ICS	International Classification for Standards. Ett internationellt klassificeringssystem för standarder.
IEC	International Electrotechnical Commission. Utarbetar internationell standard inom området el.
ISO	International Organization for Standardization. Utarbetar internationell standard inom alla områden utom telekommunikation och elteknik.
ITS	Informationstekniska standardiseringen. Utarbetar och bevakar standardisering inom informationsteknik. En av de svenska medlemmarna i ETSI, European Telecommunications Standards Institute.
ITU	International Telecommunication Union. Utarbetar internationella standarder inom radio och telekommunikation.
IWA	ISO Workshop Agreement. Tekniskt dokument utarbetad av ISO-organiserad arbetsgrupp.
Konsoliderad	En konsoliderad standard har sitt tillägg inarbetat och ersätter tidigare utgåva.
PAS	Publicly Available Specifications. Tekniska dokument inom IEC och ISO.
prEN	Förslag till Europastandard från CEN, CENELEC eller ETSI.
SEK	SEK Svensk Elstandard. Svarar för standardiseringen inom området el i Sverige. Svensk medlem i CENELEC och IEC.
SIS	SIS, Swedish Standards Institute. Svensk medlem i CEN och ISO.
SIS-TR	Technical Report. Teknisk rapport som beskriver resultat av undersökningar eller andra studier.
SIS-TS	Technical Specification. Teknisk specifikation som anger tekniska krav som ska uppfyllas av en produkt, process eller tjänst.
SIS-WA	SIS Workshop Agreement. Överenskommelse som ger regler, riktlinjer eller kännetecken för aktiviteter eller deras resultat.
SIS-WS	SIS Workshop. Standardiseringsprojekt med syfte att snabbt ta fram SIS Workshop Agreement.
SS	Svensk standard. Fastställs av SIS, SEK eller ITS. SS ingår som första led i beteckningen för svensk standard fastställd efter 1 januari 1978.
SS/T1	Tillägg 1 till svensk standard.
SS-EN	Europastandard fastställd som svensk standard.
SS-EN/AC	Rättelse till Europastandard fastställd som svensk standard.
SS-EN/A1	Tillägg 1 till Europastandard fastställd som svensk standard.
SS-EN ISO	Internationell standard från ISO som blivit Europastandard och fastställd som svensk standard.
SS-EN ISO/AC	Rättelse till standard från ISO som blivit Europastandard och fastställd som svensk standard.
SS-EN ISO/A1	Tillägg 1 till standard från ISO som blivit Europastandard och fastställd som svensk standard.
SS-EN ISO/IEC	Internationell standard från ISO/IEC som blivit Europastandard och fastställd som svensk standard.
SS-IEC	IEC-standard fastställd som svensk standard.
SS-ISO	ISO-standard fastställd som svensk standard.
SS-ISO Amd 1	ISO-standard fastställd som svensk standard. Tillägg 1.
SS-ISO/Cor 1	ISO-standard fastställd som svensk standard. Rättelse 1.
WD	Working Draft. Förslag till internationell standard eller Europastandard utarbetade i WG.
WG	Working Group. Arbetsgrupp tillsatt av t. ex. en internationell kommitté.
WI	Work Item. Ärende, en avgränsad arbetsuppgift som avser att resultera i en standard.

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Glossary

Here are a number of the abbreviations/acronyms frequently used in standardisation contexts, with brief explanations. The abbreviations are in alphabetical order.

CEN	European Committee for Standardization (Comité Européen de Normalisation). Develops European standards for areas not covered by CENELEC.
CENELEC	European Committee for Electrotechnical Standardization (Comité Européen de Normalisation Electro-technique). Develops European standards in the electricity sector.
CWA	CEN/CENELEC Workshop Agreement. Technical document developed by CEN- and CENELEC-organised working group.
Consolidated	A consolidated standard incorporates its supplement and replaces previous editions.
EN	European standard from CEN/CENELEC.
ETSI	European Telecommunications Standards Institute. Develop European standards in the telecommunications field.
FDIS	Final Draft International Standard. Final proposal for global standard from IEC or ISO.
ICS	International Classification for Standards. An international classification system for standards.
IEC	International Electrotechnical Commission. Develops global standards in the electricity sector.
ISO	International Organization for Standardization. Develops global standards in all areas except telecommunications and electrical technology.
ITS	ITS Information Technology Standardisation. Develops and monitors standardisation in information technology. A Swedish member of the ETSI, European Telecommunications Standards Institute.
ITU	International Telecommunication Union. Develops global standards in radio and telecommunications.
IWA	ISO Workshop Agreement. Technical document developed by ISO-organised working group.
PAS	Publicly Available Specifications. Technical IEC and ISO documents.
prEN	Draft European standards from CEN, CENELEC or ETSI.
SEK	SEK Svensk Elstandard. Develops and monitors standardisation in the electrotechnical sector. Swedish member of the CENELEC and IEC.
SIS	SIS, Swedish Standards Institute. Swedish member of CEN and ISO.
SIS-TR	Technical Report. Technical report describing the results of investigations or other studies.
SIS-TS	Technical Specification. Technical specification describing what requirements a product, process or service must fulfil.
SIS-WA	SIS Workshop Agreement. An agreement setting out rules, guidelines or criteria for activities or their results.
SIS-WS	SIS Workshop. Standardisation project aimed at rapidly developing an SIS Workshop Agreement.
SS	Swedish standard. Established by SIS, SEK or ITS. SS is the first step in all standard classifications to emerge since 1 January 1978.
SS/T1	Supplement 1 to a Swedish standard.
SS-EN	European standard established as Swedish standard.
SS-EN/AC	Correction to the European standard established as Swedish standard.
SS-EN/A1	Supplement to the European standard established as Swedish standard.
SS-EN ISO	International and European standard established as Swedish standard.
SS-EN ISO/AC	Correction to the international and European standard established as Swedish standard.
SS-EN ISO/A1	Supplement to the international and European standard established as Swedish standard.
SS-EN ISO/IEC	International and European standard established as Swedish standard.
SS-IEC	IEC standard established as Swedish standard.
SS-ISO	ISO standard established as Swedish standard.
SS-ISO Amd 1	ISO standard established as Swedish standard, with Amendment 1.
SS-ISO/Cor 1	ISO standard established as Swedish standard, with Correction 1.
WD	Working Draft. Proposed international or European standard developed by the Working Group.
WG	Working Group. Appointed by an international committee or some other body.
WI	Work Item. A delimited task designed to result in a standard.

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